IV Infusions Practice 4

Module 3 - Intravenous Infusions · Daily practice sheet · No date

Name _____ Date _____

The therapeutic plasma level of a drug is 4 to 15 mg/L. You would like to immediately attain a steady-state concentration level of 10 mg/L and maintain this concentration over 12 hours. The drug has a half-life of 6 hours and renal and metabolic clearances of 5 and 6.55 L/hr, respectively (total body clearance = renal clearance + metabolic clearance). The patient's body weight is 65 kg.

a Determine an IV bolus loading dose and infusion rate to achieve the desired steady-state plasma level of 10 mg/L. (Answer here: the infusion rate.)

a Determine an IV bolus loading dose and infusion rate to achieve the desired steady-state plasma level of 10 mg/L. (Answer here: the IV bolus loading dose.)

b Calculate the volume and concentration of the infusion solution if the rate of input of the solution is 1 mL/min. (Answer here: the volume.)

b Calculate the volume and concentration of the infusion solution if the rate of input of the solution is 1 mL/min. (Answer here: the concentration.)

c What is the apparent volume of distribution?

d If there were no loading dose, how much time would be required to reach the 5 mg/L level?

e If there were no loading dose, how much time would be required to reach 80% of the steady-state plasma concentration?

f Calculate the concentration of drug in the plasma at 1, 6 and 12 hours following the cessation of the infusion at the steady-state level of 10 mg/L. (Answer here: 1 hour after.)

f Calculate the concentration of drug in the plasma at 1, 6 and 12 hours following the cessation of the infusion at the steady-state level of 10 mg/L. (Answer here: 6 hours after.)

f Calculate the concentration of drug in the plasma at 1, 6 and 12 hours following the cessation of the infusion at the steady-state level of 10 mg/L. (Answer here: 12 hours after.)

g If the patient develops acute renal failure and the renal clearance is reduced to 2 L/hr, what IV bolus loading dose and infusion rate would you recommend to achieve the steady-state level of 10 mg/L? Assume metabolic clearance and the apparent volume of distribution remain constant. (Answer here: the infusion rate.)

g If the patient develops acute renal failure and the renal clearance is reduced to 2 L/hr, what IV bolus loading dose and infusion rate would you recommend to achieve the steady-state level of 10 mg/L? Assume metabolic clearance and the apparent volume of distribution remain constant. (Answer here: the IV bolus loading dose.)